

OTEPIC

MULCHING



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TRAINING MANUAL HANDBOOK

MULCHING

Mulching is the process of covering the topsoil with plant material such as leaves, grass, twigs, crop residues, straw etc. A mulch cover enhances the activity of soil organisms such as earthworms.

A mulch is a layer of material applied to the surface of soil. Reasons for applying mulch include conservation of soil moisture, improving fertility and health of the soil, reducing weed growth, and enhancing the visual appeal of the area.

A mulch is usually, but not exclusively, organic in nature. It may be permanent (e.g. plastic sheeting) or temporary (e.g. bark chips). It may be applied to bare soil or around existing plants. Mulches of manure and compost will be incorporated naturally into the soil by the activity of worms and other organisms. The process is used both in commercial crop production and in gardening, and when applied correctly, can improve soil productivity.



Advantages of mulching

Mulching, the practice of covering the soil with organic or inorganic materials, offers numerous benefits for gardens and agricultural systems. **Here are the key advantages of garden mulching:**

Moisture Retention: Reduces Evaporation: Mulch acts as a barrier that slows down the evaporation of water from the soil, helping maintain consistent soil moisture levels, especially during hot or dry periods.

Decreases Watering Frequency: By conserving moisture, mulching reduces the need for frequent watering, saving time, labor, and water resources.

Weed Control: Suppresses Weed Growth: Mulch blocks sunlight from reaching weed seeds, preventing them from germinating and growing. This helps reduce the need for chemical herbicides and manual weeding.

Reduces Weed Competition: By preventing weeds from establishing, mulch minimizes competition for nutrients, water, and light, allowing desired plants to grow more effectively.

Soil Temperature Regulation: Insulates the Soil: Mulch helps regulate soil temperatures by keeping the soil cooler in summer and warmer in winter. This temperature stability promotes better root growth and prevents temperature extremes from stressing plants.

Prevents Soil Freezing: In colder climates, mulch can act as insulation, protecting plant roots from freezing during winter.

Improved Soil Fertility: Adds Organic Matter: Organic mulches like straw, leaves, or grass clippings gradually decompose, adding valuable organic matter to the soil. This improves soil structure, fertility, and nutrient availability over time.

Enhances Soil Microbial Activity: As mulch breaks down, it stimulates the growth of beneficial soil microbes that enhance soil health and nutrient cycling.

Prevention of Soil Erosion: Reduces Soil Splashing: Mulch helps protect the soil surface from the impact of heavy rain, reducing soil erosion and the loss of topsoil.

Stabilizes the Soil Surface: By covering the soil, mulch reduces wind erosion and the movement of soil particles, keeping the soil in place and preserving its structure.

Improved Aesthetic Appeal: Enhances Garden Appearance: Mulch provides a neat, uniform look to garden beds, improving the overall aesthetic of your garden or landscape. It also complements plant colors and textures.

Reduces Bare Soil: Mulching covers unsightly bare soil, reducing the appearance of weeds and creating a clean, well-maintained look.

Reduction of Soil Compaction: Prevents Soil Disturbance: Mulch helps protect the soil from being compacted by foot traffic or heavy rainfall. This is particularly important in garden beds where soil compaction can hinder root growth and water infiltration.

Promotes Airflow: By preventing soil crusting, mulch encourages better airflow to plant roots, improving oxygen availability and overall plant health.

Pest Control: Discourages Pest Habitats: Certain mulches, such as wood chips, can deter pests like slugs, snails, and certain insects by creating an environment that is unfavorable for them to thrive.

Reduces Disease Spread: By preventing soil from splashing onto plant leaves during rainfall, mulch can reduce the spread of soil-borne diseases, such as fungal infections and blight.

Enhances Plant Growth: Promotes Stronger Roots: Consistent moisture levels and improved soil structure support the development of deeper, healthier roots, leading to stronger, more resilient plants.

Increased Plant Productivity: By improving soil conditions and reducing stress on plants, mulch can help increase yields in vegetable and fruit gardens.

Carbon Sequestration: Carbon Storage: Organic mulches such as leaves, straw, or wood chips slowly decompose, sequestering carbon in the soil. This contributes to mitigating climate change by storing carbon in the ground.

Reduces Labor and Maintenance: Less Frequent Weeding: By preventing weeds from growing, mulching reduces the time and effort required for weeding, allowing gardeners to focus on other tasks.

Less Need for Fertilizer: As mulch breaks down and enriches the soil, it can reduce the need for supplemental fertilizers, lowering overall input costs.

MATERIALS USED

Materials used as mulches vary and depend on a number of factors. You have to take into consideration the availability, the cost, the appearance and the effect it has on the soil, including chemical reactions and pH, durability, combustibility, rate of decomposition,

A variety of materials are used as mulch:

- Organic residues: grass clippings, leaves, hay, straw, kitchen scraps, comfrey,^[15] shredded bark, whole bark nuggets, sawdust, etc
- Compost: fully composted materials (humus) are used to avoid possible phytotoxicity problems. Materials that are free of seeds are ideally used, to prevent weeds being introduced by the mulch.
- Rubber mulch: made from recycled tire rubber.
- Plastic mulch: crops grow through slits or holes in thin plastic sheeting.
- Rock and gravel can also be used as a mulch.^[17] In cooler climates the heat retained by rocks may extend the growing season.

Organic mulches

Organic mulches decay over time and are temporary. The way a particular organic mulch decomposes and reacts to wetting by rain and dew affects its usefulness.

NOTE: Some mulches such as straw, peat, sawdust and other wood products may for a while negatively affect plant growth because of their wide carbon to nitrogen ratio because bacteria and fungi that decompose the materials remove nitrogen from the surrounding soil for growth.

Organic mulches can mat down, forming a barrier that blocks water and air flow between the soil and the atmosphere.

Commonly available organic mulches include:

Leaves

Leaves from deciduous trees, which drop their foliage in the autumn/fall. They tend to be dry and blow around in the wind, so are often chopped or shredded before application. As they decompose, they adhere to each other but also allow water and moisture to seep down to the soil surface. Thick layers of entire leaves, especially of maples and oaks, can form a soggy mat in winter and spring which can impede the new growth of lawn grass and other plants.

Dry leaves are used as winter mulches to protect plants from freezing and thawing in areas with cold winters; they are normally removed during spring.



Grass Clippings

Grass clippings, from mowed lawns are sometimes collected and used elsewhere as mulch. Grass clippings are dense and tend to mat down, so are mixed with tree leaves or rough compost to provide aeration and to facilitate their decomposition without smelly putrefaction. Rotting fresh grass clippings can damage plants; their rotting often produces a damaging buildup of trapped heat.

Grass clippings are often dried thoroughly before application, which militates against rapid decomposition and excessive heat generation. Fresh green grass clippings are relatively high in nitrate content, and when used as a mulch, much of the nitrate is returned to the soil, conversely the routine removal of grass clippings from the lawn results in nitrogen deficiency for the lawn.



Wood Chips

Wood chips are a byproduct of the pruning of trees by arborists, utilities and parks; they are used to dispose of bulky waste. Tree branches and large stems are rather coarse after chipping and tend to be used as a mulch at least three inches thick. The chips are used to conserve soil moisture, moderate soil temperature and suppress weed growth. Wood chip mulches on the top of the soil increase nutrient levels in soils and associated plant foliage, contrary to the myth that wood chip mulch tie up nitrogen.^{[22][23][24]} Wood chips are most often used under trees and shrubs. When used around soft stemmed plants, an un-mulched zone is left around the plant stems to prevent stem rot or other possible diseases. They are often used to mulch trails, because they are readily produced with little additional cost outside of the normal disposal cost of tree maintenance. Wood chips come in various colors.

Bark Chips

Bark chips of various grades are produced from the outer corky bark layer of timber trees. Sizes vary from thin shredded strands to large coarse blocks. The finer types are very attractive but have a large exposed surface area that leads to quicker decay. Layers two or three inches deep are

usually used, bark is relatively inert and its decay does not demand soil nitrates. Bark chips are also available in various colors.

Straw Mulch / Field Hay / Salt Hay

Permaculture garden with a fruit tree, herbs, flowers and vegetables mulched with hay

Straw mulch or field hay or salt hay are lightweight and normally sold in compressed bales. They have an unkempt look and are used in vegetable gardens and as a winter covering. They are biodegradable and neutral in pH. They have good moisture retention and weed controlling properties but also are more likely to be contaminated with weed seeds. Salt hay is less likely to have weed seeds than field hay. Straw mulch is also available in various colors.

Biodegradable mulch

Biodegradable mulches are made out of plant starches and sugars or polyester fibers. These starches can come from plants such as wheat and maize.^[27] These mulch films may be a bit more permeable allowing more water into the soil. This mulch can prevent soil erosion, reduce weeding, conserve soil moisture, and increase temperature of the soil.^[28] Ultimately this can reduce the amount of herbicides used and manual labor farmers may have to do throughout the growing season. At the end of the season these mulches will start to break down from heat. Microorganisms in the soil break down the mulch into two components, water and carbon dioxide, leaving no toxic residues behind.^[27] This source of mulch requires less manual labor since it does not need to be removed at the end of the season and can actually be tilled into the soil.^[27] With this mulch it is important to take into consideration that it is much more delicate than other kinds. It should be placed on a day which is not too hot and with less tension than other synthetic mulches.^[27] These also can be placed by machine or hand but it is ideal to have a more starchy mulch that will allow it to stick to the soil better.

Cardboard / Newspaper

Cardboard or newspaper can be used as semi-organic mulches. These are best used as a base layer upon which a heavier mulch such as compost is placed to prevent the lighter cardboard/newspaper layer from blowing away. By incorporating a layer of cardboard/newspaper into a mulch, the quantity of heavier mulch can be reduced, whilst improving the weed suppressant and moisture retaining properties of the mulch.^[13] However, additional labour is expended when planting through a mulch containing a cardboard/newspaper layer, as holes must be cut for each plant. Sowing seed through mulches containing a cardboard/newspaper layer is

impractical. Application of newspaper mulch in windy weather can be facilitated by briefly pre-soaking the newspaper in water to increase its weight.

Conclusion

Garden mulching provides multiple benefits, including moisture retention, weed suppression, temperature regulation, soil fertility improvement, and enhanced plant growth. It is a simple yet effective technique that contributes to healthier, more productive gardens while promoting sustainability and reducing labor. Whether using organic or inorganic mulches, the practice is an essential tool in maintaining a thriving, low-maintenance garden.

Mulching Picture Gallery

